SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I [redacted] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1) Scenario: Event Communication security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing a message "Meet at Noon" is transformed into a encoded version using a fixed shifted pattern.

a) Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Plain text: MEET AT NOON

In the caesar cipher, each letter is shifted four positions forward in the alphabet.

Plain text:

M E E T A T N O O N

Cipher text:

Q I I X E X R S S R

Encrypted message: QIIX EX RSSR

b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

The receiver shifts each letter of the cipher text four positions backwards

For example, Q becomes M, I becomes E, and X becomes T.

After applying the reverse shift, the original message is recovered.

Decrypted message: MEET AT NOON.

2) Atbash cipher

a) Deduce the most likely original message.

Ciphertext: GSRH RH Z HVXIVG

The Atbash cipher replaces each letter with its reverse counterpart in the alphabet.

Decrypted message: THIS IS A SECRET

b) Justify your answer by analyzing repeating patterns, letter structures or known cipher characteristics.

The Atbash cipher uses the following letter substitutions:

* A ↔ Z
* B ↔ Y
* C ↔ X
* D ↔ W

Applying this rule to the ciphertext produces THIS IS A SECRET. The original word lengths and spaces are preserved and the decrypted sentence is meaningful.

Therefore, the message was encrypted using the Atbash cipher.

3) Vigenere cipher

a) Demonstrate how the encryption process transforms the message step-by-step.

Plaintext: HELLO

Keyword: KEY

Repeat the keyword to match the plaintext length: KEY KE

Encryption formula:-

$$C = (P + K) \bmod 26$$

Here A=0, B=1, C=2, ... Z=25.

Plaintext	keyword	calculation	Cipher text
H(7)	K(10)	(7+10) mod 26 = 17	R
E(4)	E(4)	(4+4) mod 26 = 8	I
L(11)	Y(24)	(11+24) mod 26 = 9	J
L(11)	K(10)	(11+10) mod 26 = 21	V
O(14)	E(4)	(14+4) mod 26 = 18	S

Encrypted message : RIJVS

b) Explain how the repeating keyword influences the ciphertext.

The keyword determines the shift applied to each plaintext letter. Since the keyword contains only three letters, it is repeated to encrypt all the letters of HELLO.

Different keyword letters produce different shifts, resulting in the ciphertext RIJVS.

b)

The receiver uses the same keyword and the reverse operation to recover the original message.

4) Image steganography.

a) Analyze how investigators could detect the presence of hidden data within the image.

Image steganography is the technique of hiding confidential information inside an image file.

Investigators can use the following methods:

1. Visual analysis: Examine the image for unusual colours, distortions, or pixel patterns.

2. Metadata analysis: check image properties for suspicious or unexpected information.

3. LSB analysis: Examine the least significant bits of pixel values for possible hidden data.

4. File analysis: Check the file size and internal structure for unexpected embedded content.

5. Steganalysis tools: use specialized tools to identify patterns that may indicate hidden information.

b) Recommend two techniques to strengthen the security of hidden communication.

1) Encryption:- Encrypt the confidential message before technique embedding in the ~~message~~ image. This prevents unauthorized users from easily understanding the extracted information.

2) Radomized data embedding: Embed the message in selected pixel locations using a secret key. This can make hidden information more difficult to detect.

Conclusion:- Combining encryption with steganography provides better protection for confidential communication.

5) Rail Fence cipher

a) Determine the most probable original message by reversing the encryption process.

Ciphertext:- WECRLTEERDSOEEFEAOCAIVDEN

The three-rail Rail Fence cipher encrypts a message by arranging its letters in a zigzag patterns across three rows and then reading the rows sequentially.

To decrypt the message:-

1- Identify the three-rail zigzag pattern.
2- Reconstruct the three rows using the ciphertext.
3- Read the letters along the zigzag path to recover the plaintext.

Original message:-

WE ARE DISCOVERED FLEE AT ONCE

b) Explain why this method is vulnerable in modern communication systems.

The Rail Fence cipher has the following weaknesses:-

1- It rearranges letters without changing their values.
2- Letter frequencies remain unchanged.
3- The number of rails can be guessed or tested.
4- It provides limited protection against cryptanalysis.
5- It does not provide adequate confidentiality for modern communication.

c) suggest one practical enhancement.

the company should replace the Rail Fence cipher with AES (Advanced Encryption Standard) a modern symmetric encryption algorithm.

AES provides stronger confidentiality when implemented with a secure encryption mode and proper key management.

Conclusion:- Replacing the Rail Fence cipher with AES-based encryption improves the security of corporate message.